Review Article

Anaesthetic considerations and challenges during awake craniotomy

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Abstract

This article summarises the key anaesthetic considerations and challenges surrounding the perioperative management of a patient undergoing awake craniotomy. The main goals include patient comfort, facilitation of patient cooperation during the critical awake phase and maintenance of optimal operating conditions. These are achieved through appropriate patient selection and preparation, familiarity with the complexity of each surgical phase and potential complications that may arise, as well as maintenance of close communication among all team members. Challenges such as loss of patient cooperation, loss of airway, intraoperative nausea and vomiting, seizures, cerebral oedema, hypertension, blood loss and use of intraoperative magnetic resonance imaging are discussed. The importance of teamwork, competence, vigilance and clear management strategies for potential complications to maximise patient outcomes is also highlighted.

Keywords: Anaesthetic considerations, awake craniotomy, challenges

INTRODUCTION

Awake craniotomy is an open intracranial surgical procedure where the patient is deliberately awake for part of the surgery. It presents a unique set of considerations and challenges to the anaesthesiologist. Underscoring these challenges lies in the requirement for good teamwork and close communication between the surgical and anaesthetic teams perioperatively for best patient outcome.

This article discusses the issues related to patient selection, surgical complexity and anaesthetic technique, all within the context of anticipated perioperative complications and challenges, and the contingency measures to manage these.

Brief overview of awake craniotomy

There is increasing evidence to suggest that awake craniotomy is associated with fewer late neurological deficits,[1] shorter hospital stay, and fewer episodes of hypotension, hypothermia and postoperative nausea and vomiting (PONV), when compared to matched procedures under general anaesthesia (GA).[2]

The main objective of anaesthesia for awake craniotomy includes allowing patient cooperation with neurological testing during parts of the surgery.

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INDICATIONS

Awake craniotomy can be performed for a variety of indications. These include the following:

1. Resection of lesions close to or within eloquent motor, cognitive or sensory cortical areas;
2. Procedures requiring electrocorticographic mapping or precise electrophysiological recordings, such as epilepsy surgery and deep brain stimulation;
3. Resection of vascular lesions, which is essential for the blood supply of functionally important territories;
4. Faster recovery and earlier discharge for minor intracranial procedures with no functional goa.l[3]

The patients are an essential part of the surgery, as they are able to give immediate feedback to the surgical team while surgery takes place on critical parts of the brain.

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PHASES OF AWAKE CRANIOTOMY

The awake craniotomy can be divided into three main phases, the preawake phase, awake phase and the postawake phase, each with its own anaesthetic considerations [Table 1].[4] The preawake phase includes pinning and positioning, intraoperative scanning to aid navigation and mapping, skin incision, craniotomy and dural opening. The awake phase consists of brain mapping, functional testing and tumour resection. Where available, surgery may be interrupted to acquire intraoperative images to show the extent and accuracy of tumour resection.

Brain mapping during the awake phase involves direct cortical electrical stimulation while the patient is awake and able to attempt to perform relevant tasks. This would allow the surgical team to determine if the stimulus disrupts execution of the task.[5] Positive mapping seeks to identify the eloquent cortical and subcortical tissues, while negative mapping identifies areas where eloquent tissue is absent. This enables resection of suspicious, yet nonfunctional cortex, allowing for more aggressive excision of the lesion while minimising surgery-induced morbidity.[6]

Finally, the postawake phase concludes the procedure with haemostasis and closure of dura, skull and skin.[7] An operating theatre that has scanning facilities may conclude the procedure with a postoperative scan to check for the extent of resection or haematoma.

Table 1. Stages of awake craniotomy.

Stage	Event	AAA technique	Monitored care with local infiltration
Preawake	Pinning, positioning	General anaesthesia	Sedation
	MRI scanning, neuronavigation		
	Skin infiltration/nerve blocks		
	Craniotomy, dural opening		
Awake	Mapping, functional testing	Awake	Sedation
	Tumour/lesion resection		
	MRI scanning if indicated		
Postawake	Haemostasis	General anaesthesia	Sedation
	Dural closure, replacement of skull bone, wound closure		
	Postoperative scanning if indicated		

AAA: awake–asleep–awake, MRI: magnetic resonance imaging

INTRAOPERATIVE TESTING

The success of an awake craniotomy procedure depends on the patient's ability to tolerate the procedure and cooperate with relevant intraoperative neurological testing. Intraoperative testing may include testing of the motor and sensory pathways, speech and language, as well as visual pathways.[4]

To elicit and interpret responses to stimuli at the relevant areas of the brain, specialised help, involving a neurologist or a speech therapist, may be required, although the anaesthetist may be called upon to help with assessing the patient when specialist assessors have not been invited to do so. The tests and assessments during surgery should be rehearsed with the patient preoperatively.

During monitoring of the motor and sensory pathways, cortical mapping can elicit or inhibit movements and sensations. The limbs, face and mouth are commonly observed for tingling, twitching or movement. The patient and the person assessing the patient should report any abnormal movement or sensation during stimulation mapping to the surgeon, so as to allow delineation of cortical and subcortical tracts. Speech and language areas cannot be safely localised based on anatomical landmarks. This often requires the aid of several specialised tests conducted by a speech therapist, including the Visual Object Naming Test and the Boston Naming Test. Bilingual patients are tested in both languages as the corresponding anatomical areas may not entirely overlap. Mapping of the cortical visual cortex and subcortical visual tracts may also be performed to reduce permanent injury in tumour resections that involve visual pathways. These assessments are demanding on patients, and a comfortable but lucid patient is necessary for accurate and consistent assessments.

ANAESTHETIC CONSIDERATIONS

Preoperatively, appropriate patient selection and preparation are crucial to the success of awake craniotomy. There are suggestions that from a neurosurgical point of view, tumour resections involving large, highly vascular tumours with the potential for significant bleeding and tumours with locations that cause significant dural pain or require uncomfortable intraoperative body positioning are better managed under GA, guided by functional magnetic resonance imaging (MRI) and intraoperative navigation.[7] Patient refusal is viewed as an absolute contraindication, while certain comorbidities have been suggested as relative contraindications because these might make awake craniotomy particularly challenging.[4,8] This will be discussed in detail later.

Patient preparation is of utmost importance to allay preoperative anxiety and facilitate intraoperative cooperation. Anaesthetic preoperative consultation is an important opportunity to provide details regarding awake craniotomy and answer outstanding questions.[8] This is best performed by the same anaesthesiologist administering the anaesthetic, so that rapport can be established early.

Intraoperatively, an important factor in the success of an awake craniotomy procedure is adequate analgesia using local anaesthesia (LA) infiltration or regional nerve blocks. This is done before pinning and is commonly achieved with a combination of lidocaine and a longer-acting

agent (bupivacaine or ropivacaine).[4] The patient's head is usually secured in a pinned head frame after LA is given. This prevents the head from moving intraoperatively, but it may severely restrict manipulation of the head and neck position during airway emergencies.[9]

The patient is usually placed in a supine, semi-lateral or lateral position during awake craniotomy.[4] The chosen position should permit surgical access to the lesion side of the brain, optimise positional endurance, as well as allow face-to-face interaction with the anaesthetic team and adequate airway access in the event of an emergency.[4,7,9,10] The anaesthesiologist's constant access to the patient's face and limbs should be considered when draping the operating field. This is accomplished by constructing a tent with full access to the patient's face.[7]

Several anaesthetic techniques have been described for awake craniotomy. It is not uncommon for the entire procedure (preawake till postawake) to be conducted wholly under local anaesthetic filtration with sedation (monitored anaesthesia care (MAC) technique). Some centres would perform the preawake and postawake phases under GA while conducting the intraoperative part under local anaesthetic with sedation (Asleep–Awake–Asleep/Awake [AAA]) technique.[11]

Regardless of the anaesthetic technique, the overriding axiom is patient safety, supporting the main goals which are to ensure an awake, comfortable and cooperative patient during the critical mapping phase, to facilitate acquisition of reliable neurophysiological monitoring signals and to ensure slack brain during resection.[5] Close communication between the surgical and anaesthetic teams should be maintained to ensure smooth transitions between the phases. In the MAC technique, spontaneous ventilation is maintained using low doses of sedative agents.

For the AAA technique, the use of higher doses results in an anaesthetic depth ranging from deep sedation to GA requiring mechanical ventilation via laryngeal mask airway or endotracheal intubation during the preawake and potentially the postawake phases. Patients typically require a lower anaesthetic depth during the postawake phase than during the preawake phase. This may be due to fatigue from the awake phase, psychological relief after lesion removal and the lower level of painful stimuli during skull closure compared to that at skull opening.[9] The benefits of the AAA technique compared to the MAC technique include prevention of accidental patient movements, better comfort and pain control for the patient. Securing the patient's airway and controlled ventilation during the first stage of awake craniotomy allow for reduction in brain swelling via hyperventilation.[11]

Postoperatively, the patient should initially be cared for in a specialised high-dependency unit or intensive care unit (ICU) familiar with neurosurgical patients. Close neurological monitoring is essential as postoperative haematomata or cerebral oedema can develop, especially in the first few hours after operating, which may require urgent treatment.[12]

ANAESTHETIC CHALLENGES DURING AWAKE CRANIOTOMY

Several factors may pose challenges to the anaesthetic management during an awake craniotomy, and in rare cases, it could result in failure to complete the procedure [Box 1].

Personality of patient

Patient collaboration and motivation are paramount for successful functional testing intraoperatively. The procedure and functional testing component of the operation may be lengthy and requires patient cooperation throughout the majority of the session. Hence, this relatively unconventional and potentially stressful procedure necessitates adequate mental preparation of the patient by the surgeon, anaesthetist, as well as speech therapist if speech domains will be tested intraoperatively. The purposes for performing the surgery awake must be explained, along with the sequence of events, expected duration and possible sources of discomfort, the associated risks and possible complications, and the means used to limit discomfort. The patient should be familiarised with the functional tests that will be performed intraoperatively. Such preparation will help the patient appreciate the procedure better, regulate his/her expectations, ameliorate any anxiety and cooperate with the procedure.[8]

Patient comorbidity

Certain comorbidities may increase the risk of perioperative complications for awake craniotomy, add to the anaesthetic challenges of the procedure and pose as relative contraindications.

Particular attention should be paid to any potentially difficult airway and the presence of obstructive sleep apnoea, both

Box 1. Anaesthetic challenges during awake craniotomy.

Patient-related
- Patient personality
- Patient comorbidity, neurological condition
- Intraoperative nausea and vomiting
- Intraoperative seizures, drop in GCS, hypoxaemia and hypercarbia
- Cerebral oedema
- Intraoperative blood pressure management

Surgical-related
- Prolonged surgery
- Intraoperative MRI scanning and its considerations and challenges
- Intraoperative blood loss

Anaesthetic-related
- Adequate analgesia and risk of LA systemic toxicity
- Intraoperative airway management/loss of airway
- Somnolence, agitation/delirium and failure of patient to cooperate

GCS: Glasgow Coma Scale, LA: local anaesthesia, MRI: magnetic resonance imaging

of which may increase the risk of airway obstruction and intraoperative loss of airway. Intraoperative haemodynamic instability may also require securing a difficult airway, especially in the presence of an exposed brain.

Patients who have difficulty lying flat as a result of significant cardiorespiratory disease may also be precluded, as respiratory distress and ventilatory failure can contribute to hypoxia and hypercarbia. This can result in agitation and drowsiness, resulting in intraoperative brain swelling.

Preexisting neurological deficits may confound intraoperative functional testing and should be documented before surgery.[3,7,8,10] Neurological problems which may make intraoperative patient cooperation and functional testing challenging include those with poor Glasgow Coma Scale (GCS) score, delirium, dementia, movement disorders and communication difficulties from dysphasia. Patients with significant anxiety, poor motivation and claustrophobia may also not be suitable candidates for an awake craniotomy.

The presence of raised intracranial pressure (ICP) preoperatively results in difficult surgery as a result of brain swelling. Other than the use of osmotic diuretics, the option of using controlled ventilation to manage this is not available during the awake phase. Patients with a history of poorly controlled seizures may be at risk of intraoperative seizures and their attendant complications (as described in the subsequent section).

Possible prolonged duration of procedure

Awake craniotomy can be potentially long, with mean durations exceeding 6 h reported.[13,14] This is further prolonged if intraoperative scanning is deployed. A retrospective study by Abaziou *et al*.[2] identified prolonged operation time after cortical mapping as an independent risk factor for major complications during awake craniotomy. More time under sedation can, itself, increase the risk of complications, especially in the post anaesthesia care unit. As the patient is tested throughout the mapping and resection phases, this process can result in significant patient fatigue.

The physical environment

Temperature management is important as the cold operating theatre environment may predispose the patient to hypothermia and shivering. These should be prevented by the use of blankets, warm air devices and setting appropriate room temperature. Bladder catheterisation should also be considered to prevent distress from a full bladder while allowing close monitoring of urine output. This should be performed with liberal amounts of lignocaine gel to minimise discomfort for the awake patient. Other sources of discomfort intraoperatively could arise from having to remain immobile on firm surfaces. This could be ameliorated to some degree with adequate padding. Other discomforts such as noise and glare of operating lights can be minimised. The prolonged surgery and patient fatigue, together with the difficult physical environment can contribute to patient agitation, with resulting loss of cooperation.[7]

Adequate analgesia

Adequate analgesia from local infiltration of LA or a regional nerve block is essential to keep the patient comfortable and cooperative during the surgery. A larger volume of LA and the well-vascularised scalps predispose to LA toxicity, and hence clinical vigilance should be paid to calculating appropriate doses and watching for the development of this complication. Clinical features that may suggest the development of LA systemic toxicity include central nervous system (manifestations, ranging from circumoral numbness and tinnitus to seizures and coma, as well as cardiovascular system manifestations such as arrhythmias, bradycardia, hypotension and cardiovascular collapse. To minimise this risk, individual nerve blocks may be preferred over wide areas of infiltration, and the addition of adrenaline minimises acute rise in plasma concentration while maximising the duration of action of the drug.[10] The lowest possible dose to achieve clinical effect should also be used, allowing room for supplementation with more LA by the surgeon if required subsequently, such that recommended toxic doses of LA are not exceeded. It should be noted that local anaesthetic toxicity is additive, whether by subsequent redosing of the same or a different local anaesthetic.[15]

Nausea and vomiting

The incidence of intraoperative nausea and vomiting during awake craniotomy has been reported to be between 0 and 9%.[16] This may potentially contribute to aspiration, loss of airway, hypoxia, reduced patient satisfaction and conversion to GA. There are risks of injury to scalp muscles and cervical spine when the patient retches while the head is pinned. Vomiting and retching also raise the ICP and may affect surgery. Routine prophylactic preoperative intravenous (IV) antiemetics have been suggested to reduce the incidence of this complication.[16,17] Also, some institutions might include patients with gastro-oesophageal reflux, uncontrolled diabetes mellitus and body mass index (BMI) >35 kg/m^2 as contraindications for having an awake craniotomy to reduce the risk of regurgitation.[17] The use of propofol, with its antiemetic effect, as the sedative agent of choice and avoidance of opioids have also been suggested to reduce the incidence of intraoperative nausea and vomiting.[18]

Intraoperative seizures, drop in GCS

The incidence of intraoperative seizures ranges between 3% and 16%, and they are most common during cortical and subcortical stimulation mapping.[6,7,9,19] The incidence is lower if the surgeon avoids stimulating an area twice in rapid succession. Continuous monitoring of electrocorticography for spikes or sharp waves within 5 s after each stimulation allows for early seizure detection.[4] A history of preoperative epilepsy may not necessarily increase the risks of intraoperative seizures.[20]

Most seizures are focal and self-limiting, but occasional conversion to generalised seizures may occur.[6] Intraoperative seizures should be controlled because of the potential for injuries such as tongue injury, musculoskeletal injuries or aspiration, and potential loss of airway with postictal drowsiness. Moreover, functional assessment of awake craniotomy cannot be accomplished when Todd paralysis, postictal aphasia or confusion results after the seizure.[21] Intraoperative seizures can also result in prolonged hospital stay.[4]

The use of ice-cold saline on brain tissue is usually sufficient to abort these seizures.[2,7] This would necessitate having ice-cold saline close by during surgery. If ineffective, small boluses of IV propofol should be administered and repeated as necessary.[4,18] The patient should then be closely monitored for seizure recurrence and airway compromise. Occasionally, benzodiazepines, antiepileptic drugs (AEDs) or resedation with airway control is required.[6,8,9,12,16] Urgent intubation and conversion to GA are recommended for prolonged seizures lasting more than 5 min.[4]

The efficacy of AEDs in intraoperative seizure prevention remains debatable. Chandra *et al*.[22] showed that AED prophylaxis did not show benefits of seizure prevention during brain tumour resection, although there is a dearth of data regarding the use of prophylaxis using levetiracetam, a more commonly used AED which has a more favourable side effect profile compared to the other AEDs, with superior seizure prevention.[22,23] It is, therefore, suggested that a combination of adequate preoperative antiepileptics (such as IV levetiracetam 500–1,500 mg) in conjunction with appropriate cortical stimulation strategies (with minimal cortical stimulation current, duration of stimulation and repetitive stimulation of specific cortical sites) can be used to reduce the incidence of intraoperative seizures.[16,24]

Intraoperative blood loss

Where lesions are vascular or have high risk of significant intraoperative bleeding, an awake craniotomy procedure may not be appropriate. Increased bleeding can also occur during the procedure as a result of hypertension and brain oedema.[18]

The chosen anaesthetic technique has also been attributed to potentially increasing incidences of intraoperative bleeding, with significantly more bleeding observed in the AAA group compared to the MAC group. This was shown to have resulted from increased incidence of hypertension, postulated to arise from the withdrawal effect of propofol and remifentanil on transition to the awake phase.[17] When unexpected massive intraoperative blood loss occurs, the anaesthetic management would include securing the airway, resuscitation with fluids and blood products, and prevention of coagulopathy as the surgical team attempts to secure haemostasis.

Excessive somnolence

Somnolence may be related to excessive sedation. This is seen with the use of the AAA technique where titration of sedation requires experience, expertise and caution.[7] The individual's susceptibility to sedative agents and opioids used during the surgery can result in oversedation. Somnolence can be managed by avoiding deep sedation during the initial and more noxious steps of the procedure. Early termination of sedative infusions and the avoidance of large doses of midazolam or longer-acting opioids have also been suggested. The use of intraoperative antiepileptic medication, especially phenytoin, has also been shown to result in somnolence during the surgery.[4,7] Other causes of intraoperative somnolence, such as hypercarbia or brain swelling, would need to be excluded.

Cerebral oedema

Intraoperative brain swelling during awake cases has been reported in 7%–14% of cases.[16] Brain swelling during awake craniotomy poses several concerns. Surgically, brain swelling may increase the risk of tearing a cerebral vessel or sinus at the time of dural opening and may be a cause for conversion of an awake case to GA.[16,18] The use of a preoperative scan for intraoperative navigation during surgery could be rendered less precise if there is significant swelling of the brain.

Contributing factors of intraoperative brain swelling may include preoperative or preexisting raised ICP, intraoperative hypercarbia and hypertension, and intractable retching or coughing. This is seen with the use of endotracheal tube in the AAA technique, causing laryngeal and tracheal stimulation. It also results in increased blood pressure and undesirable brain swelling. This is also seen when sedation is stopped during various phases of surgery.[7]

Restlessness of the patient may also result in inadvertent Valsalva manoeuvres or hypertension, which might cause imperfect haemostasis or further brain swelling.[24] Hence, by allowing the patient to wake up fully, obtaining full cooperation, especially before dural opening and cortical exposure, may prevent refractory brain swelling.[16]

The contribution of higher carbon dioxide levels to brain swelling cannot be underestimated. Two retrospective studies demonstrated that $PaCO_2$ levels were significantly higher in patients undergoing awake craniotomy with MAC compared to AAA,[17] as well as in patients undergoing craniotomy under AAA compared to GA.[18] Control of hypercarbia to manage brain swelling during the awake phase may no longer be available because of the inability to institute ventilatory measures for hypercarbia and facilitation of brain relaxation, short of asking the patient to voluntarily control their minute ventilation. However, despite higher CO_2 levels during awake craniotomy, these retrospective studies reported no increased incidence of brain oedema.[17,18]

Blood pressure management

Hypertension is a common occurrence in awake craniotomy. It may potentially contribute to intraoperative brain swelling and bleeding.[18,24] It is most commonly secondary to pain, agitation,

anxiety or discomfort from lying still for prolonged periods or occasionally occurs due to other causes such as hypoxia and hypercapnia. Treatment should focus on managing the underlying cause. Antihypertensives such as labetalol or esmolol may sometimes be necessary for blood pressure control and should be readily available.[4] Good communication between the anaesthesia and surgical teams regarding patient discomfort and haemodynamic concerns has been suggested to minimise the incidence of intraoperative hypertension. This allows the neurosurgeon to pause from proceeding with the operation, when needed, until the underlying issue has been addressed.[16]

Emergence agitation and delirium

This may occur especially when the preawake phase is performed with GA or deep sedation. Contributing factors include older age, pain, disorientation, inappropriate use of naloxone, flumazenil, neostigmine or atropine, hypoxaemia, hypercapnia, urethral stimulation and bladder distention.[4] When agitation and delirium occur in a patient whose head is pinned and secured to the table and with an exposed brain, unwanted movements may expose the patient to serious injuries, and the loss of cooperation may preclude subsequent functional testing.

Another potentially modifiable cause for agitation and delirium is the anaesthetic depth, with deeper anaesthesia associated with a higher incidence.[9,25] There is no consensus on the best approach to managing this. Reinduction of anaesthesia with titrated propofol boluses, dexmedetomidine boluses, and by using low-dose remifentanil infusion, droperidol or haloperidol before attempting reawakening are all strategies that have been suggested.[4]

Intraprocedural airway management

Perhaps one of the greatest challenges that an anaesthetist could face during awake craniotomy is securing the patient's airway emergently. This may occur with seizures, inability to tolerate the procedure, agitation or a drop in GCS.[21] The anaesthesiologist must have a strategy to expeditiously manage the patient's airway and to convert the procedure to be performed under GA if the need arises. Hypoxia and desaturation episodes occur more commonly during MAC with an unsecured airway and are commonly due to oversedation.[10]

The incidence of airway obstruction during an awake craniotomy ranges from 7% to 16%, necessitating jaw lift, or additional supplemental oxygen via a nasal airway, use of mask ventilation, even intubation for a definitive airway.[16] Although there is rarely a need for emergency airway control, there should be an emergency airway control plan, and a full range of airway equipment, including laryngeal mask airway, video laryngoscope and endotracheal tube, should be immediately available throughout the procedure.[8,18]

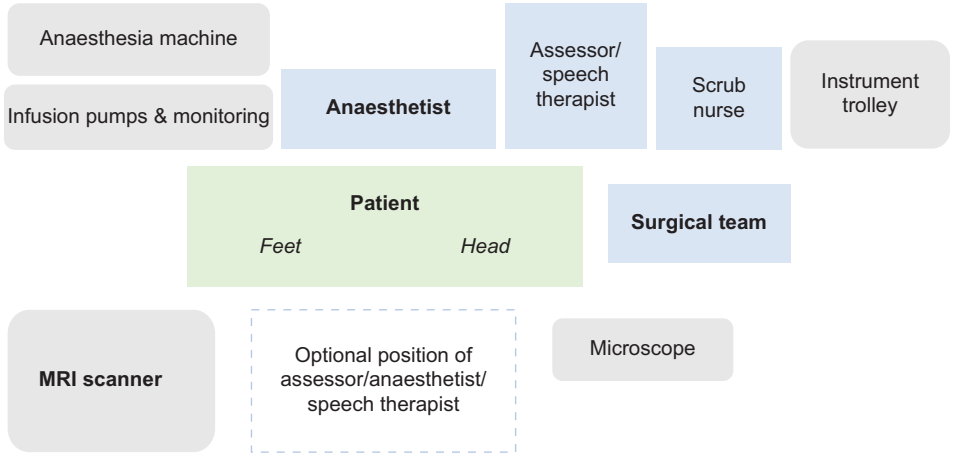
During an obstructed airway situation, an emergency or crisis situation should be announced, so that the surgical and nursing teams are alerted and skilled assistance is called for. A rapid differential diagnosis is needed to identify the cause. Mask ventilation of the patient with 100% inspired oxygen, facilitated with a jaw thrust, oral or nasal airway, may be required. Sedatives have to be stopped. In consultation with the surgical team, the patient could be placed supine, the craniotomy wound protected and the head clamp released. This facilitates laryngeal mask airway insertion and endotracheal intubation with skilled help and possible adjuncts such as videolaryngoscopy, since both could be very difficult with the patient's head immobilised by pins and clamps.[9,12]

Challenges resulting from the use of iMRI

Awake craniotomy may be combined with the use of intraoperative MRI (iMRI) to maximise the extent of accurate tumour resection by identification of tumour planes and residual tumour through repeated imaging during surgery. This helps preserve critical neurological function and optimise long-term neuro-oncological outcomes and quality of life.[8,14,26] However, the need for intraoperative scanning prolongs the total duration of procedure.

Anaesthetic management in the iMRI suite provides several unique challenges for anaesthetists as a result of the MRI environment [Figure 1]. The latest practice advisory from the American Society of Anesthesiologists highlights the challenges of anaesthetic administration and monitoring capabilities in magnetically active environments, due to hazards from static/gradient magnetic fields and radiofrequency energy.[27] Staff should be cognisant of the 5-G line within the MRI environment, beyond which ferromagnetic objects could potentially become dangerous projectiles. Precautions should also be undertaken to prevent heating and burns to the patient, including care in arrangement of monitoring cables to prevent closed loops and cross points, use of insulating material to insulate the patient from cables and avoidance of skin-to-skin contact. During the period of scanning, the patient is usually sedated for the duration of the scan, which typically lasts about 30–45 min.

As access to the patient may be limited during scanning, relevant equipment and monitoring displays should be visible at all times, with consideration for extension of oxygen source and fluid lines. During intraoperative scanning, direct visualisation of chest movement or auscultation may be difficult due to positioning within the scanner and room lighting. Also, immediate access to the patient's airway is restricted once the receiver coil has been placed over the patient's face. This adds to the claustrophobia from being within the magnetic bore. During scanning, concurrent use of capnography and pulse oximetry can help detect early onset of respiratory depression or respiratory arrest from oversedation or neurological changes, especially when desaturation lags significantly behind hypoventilation.[28]



iMRI = intraoperative magnetic resonance imaging

Figure 1: Set-up of awake craniotomy in the iMRI operating theatre.

TEAMWORK AND COMPETENCE OF TEAM MEMBERS

A certain confidence level and competency are expected of the team to facilitate good teamwork and making of sound decisions, especially when complications arise. However, the relative scarcity of this unique surgery may pose challenges with staff teamwork and competence. Hence, ongoing training and audit are required to provide the best patient care.

All staff involved in the procedure must be familiar with the intraoperative set-up and the unique considerations during surgery. The less-visible heroes would include the MRI radiographer and the healthcare assistant, who are instrumental in operating the MRI-compatible operating table and its fixtures. A spirit of consultation and collaboration among the staff managing the patient throughout the perioperative and intraoperative journey is essential. The personnel involved must include the neurologist, neurointensivist, neuroradiologist, neurophysiologist, speech therapists and neurosurgical nurses.

CONCLUSION

Awake craniotomy is a procedure that benefits a select group of patients, primarily in reduction of neurological deficits while maximising tumour resection. The key is to facilitate patient cooperation during the critical awake phase, while maintaining optimal operating conditions throughout the surgery. This article has described the main anaesthetic considerations and challenges, as well as how best to prevent and manage them. It is necessary to reiterate the importance of teamwork and competence, vigilance for potential complications and having clear management strategies when these arise.

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Conflicts of interest

There are no conflicts of interest.

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